MIL-P-46500 (Ord)
3 March 1961

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M78
PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of percussion primer designated as Primer, Percussion, M78.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-10025 - Packaging and Marking for Domestic Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8847521 - Plug
8847522 - Holder
8847523 - Head
8847524 - Closure

FSC 1390

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-(See 6.2) - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. - The parts shall comply with all requirements specified on Drawings (dwgs.) 8847521, 8847522, 8847523 and 8847524 as applicable, and with all requirements specified in applicable specifications.

3.3 Workmanship. - All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. - The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling

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procedures, methods of inspection (gages, measuring and testing equipment), and provisions for the control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor, when such material is covered by referenced Government specification.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

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4.1.3.1. Surveillance. - Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection. - Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records, Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies. - Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operation, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Preproduction inspection.

4.2.1 Submission. - The contractor shall submit 25 of each part which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. The preproduction quantity shall be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, or specification, such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Each assembly shall be inspected for all the inspections specified in 4.3 of this specification.

4.2.3 Rejection. - If any assembly fails to comply with the applicable requirements, the preproduction quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new preproduction quantity, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary, until such time as a quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

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4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of parts produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the lot number.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristic listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.3.

4.3.2.1 Holder (see dwg. 8847522).

Categories	Defects	Method of inspection
Critical:		
1.	Length from primer end to bottom surface of shoulder, maximum (max.)	Gage
2.	Depth of primer cavity, minimum (min.)	Gage
Major:	AQL 0.40 percent	
101.	Evidence of poor workmanship (see 3.3)	Visual
102.	Coating missing in black powder cavity	Visual
103.	Flash hole missing or completely obstructed	Visual
Minor:	AQL 4.00 percent	
201.	Pitch diameter of thread	Gage
202.	Major diameter of thread	Gage
203.	Concentricity of primer cavity with pitch diameter	Gage
204.	Diameter of flash hole	Gage
205.	Diameter of primer cavity	Gage
206.	Depth of primer cavity, max	Gage
207.	Depth of black powder cavity	Gage
208.	Depth of closure cavity	Gage
209.	Diameter of black powder cavity	Gage
210.	Diameter of thread undercut, min	Gage
211.	Outside diameter	Gage

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212.	Length between wrench flats	Gage
213.	Length from primer end to crimping lip	Gage
214.	Length from primer, end to shoulder, min	Gage
215.	Length of thread pilot	Gage
216.	Diameter and profile of closure disc cavity	Gage
217.	Height of crimping lip	Gage
218.	Crimping lip torn, ragged or distorted	Visual
219.	Finish improper	Visual
220.	Chamfer missing	Visual

4.3.2.2 Head (see dwg. 8847523).

Categories		Defects	Method of inspection
Critical:			
	1.	Depth of holder cavity, min	Gage
	2.	Depth to bottom of plug cavity counterbore, min	Gage
Major:		None defined AQL	
Minor:		AQL 4.00 percent	
	201.	Pitch diameter of external thread	Gage
	202.	Major diameter of external thread	Gage
	203.	Pitch diameter of internal thread	Gage
	204.	Minor diameter of internal thread, max	Gage
	205.	Thickness of flange	Gage
	206.	Depth to bottom of plug cavity counterbore, max	Gage
	207.	Total length	Gage
	208.	Diameter of flange	Gage
	209.	Outside diameter under flange	Gage
	210.	Length of effective internal thread, min	Gage
	211.	Diameter of plug cavity hole	Gage
	212.	Diameter of plug cavity counterbore	Gage
	213.	Location and min. diameter of wrench holes	Gage
	214.	Length of effective external thread, min	Gage
	215.	Evidence of poor workmanship (see 3.3)	Visual

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4.3.2.3 Plug (see dwg. 8847521).

Categories	Defects	Method of inspection
Critical:		
1.	Length of firing point and flange, max	Gage
Major:	AQL 0.25 percent	
101.	Profile and diameter of firing point	Gage
Minor:	AQL 2.50 percent	
201.	Diameter of pilot	Gage
202.	Diameter of flange	Gage
203.	Length of firing point	Gage
204.	Radii missing	Visual
205.	Evidence of poor workmanship (see 3.3)	Visual

4.3.2.4 Closure (see dwg. 8847524).

Categories	Defects	Method of inspection
Critical: None defined		
Major:	AQL 0.25 percent	
101.	Disc punctured, torn or otherwise mutilated to the extent that use is impaired	Visual
Minor:	AQL 1.50 percent	
201.	Diameter	Gage
202.	Thickness	Gage
203.	Evidence of poor workmanship (see 3.3)	Visual

4.3.3 Inspection equipment. - For the performance of the examinations specified in 4.3, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment (see 6.2). When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level C.-The parts shall be packed and marked in accordance with Specification MIL-P-10025.

5.2 Data cards.-Data card information shall be as specified in Specificatio MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Inspection equipment.-Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, to determine availability of inspection equipment designs.

Notice.-When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permissio to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing activity:
Army-Ordnance Corps

GPO

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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